

Azure NOTES

1. Azure Subscription

An **Azure Subscription** is your root billing and administrative account in Azure. Think of it like your home electric bill account—everything you turn on (VMs, databases, storage) gets billed to this account. Without a subscription, you cannot build anything in Azure.

1. **Quota Hard & Soft Limits:** Subscriptions have regional limits (e.g., a limit on maximum vCPU cores in East US). If your pipeline deployment fails unexpectedly, it's often because you hit a subscription core quota, which requires a support ticket to increase.
2. **Management Groups Hierarchy:** Subscriptions aren't the top tier! **Management Groups** sit *above* subscriptions to apply governance, policies, and compliance rules across multiple subscriptions.
3. **Resource Migration Limits:** While you can move resources between subscriptions, **not all services support cross-subscription migration** (e.g., certain Key Vault configurations or connected integrations must be recreated).
4. **Subscription Disablement:** If a payment fails or spending limit is hit on a free account, Azure disables the subscription, putting resources into a read-only state before eventual deletion.

2. Resource Group (RG)

A **Resource Group** is a logical container used to group related Azure resources together. Imagine you are building an e-commerce website called ShoppingApp. You put its database, server, storage, and security keys into one folder (RG-ShoppingApp). When the project ends, deleting the folder deletes everything inside in one click.

1. **Resource Groups Cannot Be Nested:** You cannot place a Resource Group inside another Resource Group (unlike standard folders on your PC).
2. **Resource Location vs. RG Location:** A Resource Group has its own location (e.g., East US), which only stores **metadata**. The actual resources *inside* the RG can reside in completely different regions (e.g., West Europe).
3. **The RG Region Outage Risk:** If the region hosting the Resource Group's metadata goes down, you may be unable to modify or manage resources in that RG, even if the resources themselves are running fine in another region.
4. **No "Recycle Bin":** Deleting an RG deletes all enclosed resources permanently. To prevent accidental catastrophic deletion, production RGs use **Resource Locks** (CanNotDelete or ReadOnly).

3. Microsoft Entra ID (Formerly Azure Active Directory)

Microsoft Entra ID is Azure's cloud identity authority. Think of Entra ID as the **Government Passport Office**. It issues passports (identities) to people, applications, and services. It handles **Authentication**—verifying *who you are* when you log in.

1. **Authentication vs. Authorization:** Entra ID verifies identity (**Authentication / AuthN**). Azure RBAC controls what that identity can do (**Authorization / AuthZ**). Authenticating successfully with Entra ID does NOT grant permission to do anything in Azure without RBAC roles.
2. **Entra Tenant vs. Azure Subscription:** An Entra Tenant is an instance of identity management representing an organization. A single Entra Tenant can manage users across **multiple subscriptions**, but a subscription can only trust **one Entra Tenant** at a time.
3. **Service Principals vs. Managed Identities:** A Service Principal is essentially an "app account" requiring manual secret/certificate management. A Managed Identity is a special wrapper around a Service Principal where Azure manages the credentials completely automatically.

4. Azure Storage Account

An **Azure Storage Account** is a cloud storage container that can hold various types of data. It serves as an entry point for four core services: **Blobs** (files, photos), **Files** (shared network drives), **Queues** (messages), and **Tables** (NoSQL data).

Azure Storage Access Tiers (Hot, Cool, Cold, Archive, Premium)

Not all data needs to be accessed with the same frequency. If you keep active website images, you need them instantly (24/7). If you keep 10-year-old tax records, you almost never open them.

Azure Blob Storage provides **Access Tiers** so you pay for the storage speed and access frequency you actually need:

- **Premium Tier:** High-speed SSD storage for intensive IOPS workloads.
- **Hot Tier:** For active data accessed frequently.
- **Cool Tier:** For data stored for at least 30 days and accessed infrequently.

- **Cold Tier:** For data stored for at least 90 days and rarely accessed.
 - **Archive Tier:** Offline storage for long-term historical records stored for at least 180 days.
1. **Global Uniqueness Rule:** Storage account names must be **globally unique across all of Microsoft Azure worldwide**, containing only 3–24 lowercase letters and numbers (e.g., mystorageacct2026). This is because it forms a public DNS URL: `https://<name>.blob.core.windows.net`.
 2. **Access Tiers & Retrieval Penalties:** Blob storage offers **Hot, Cool, Cold**, and **Archive** tiers.
 - *Archive* tier is extremely cheap for storage, but data is offline. Rehydrating data from Archive to Hot takes *hours* and carries heavy financial penalties if accessed before 180 days.
 3. **Egress Cost Gotcha:** Ingesting data into Azure (Ingress) is free. Moving data *out* of an Azure region or out to the internet (Egress) incurs bandwidth costs.
 4. **Storage Redundancy Tiers:**
 - **LRS (Locally Redundant):** 3 copies in 1 data center.
 - **ZRS (Zone Redundant):** 3 copies across 3 availability zones in 1 region.
 - **GRS (Geo-Redundant):** Cross-region replication to guard against primary region disaster.

5. Azure Data Lake Storage Gen2 (ADLS Gen2)

ADLS Gen2 is specialized cloud storage optimized for massive big data analytics. Standard Blob storage treats folders like fake text labels (folder1/file.txt), whereas ADLS Gen2 provides real directory trees just like your computer.

1. **One-Way Decision (HNS Toggle):** Enabling **Hierarchical Namespace (HNS)** turns standard Blob storage into ADLS Gen2. *This cannot be disabled after creation.* You must recreate the storage account to change it.
2. **RBAC vs. ACL Permissions Conflict:** ADLS Gen2 supports both Azure RBAC and POSIX-style ACLs (Access Control Lists).
 - **Rule: RBAC takes precedence.** If a user has RBAC Storage Blob Data Contributor, ACLs cannot restrict them. ACLs are only evaluated if no granting RBAC role exists at higher scopes.

6. Managed Identity

In traditional systems, code needs passwords or API keys stored in configuration files to connect to databases. If an engineer accidentally pushes code to GitHub, credentials leak.

A **Managed Identity** gives Azure services an automatically managed identity card inside Microsoft Entra ID. Services talk to each other without developers ever seeing or managing passwords.

1. **System-Assigned vs. User-Assigned Lifecycle:**
 - **System-Assigned:** Tied 1:1 with a specific resource instance. Deleting the VM/Databricks workspace **deletes the identity**.
 - **User-Assigned:** Independent Azure resource. Can be created beforehand and assigned to *multiple* resources simultaneously.
2. **Under-the-Hood IMDS Mechanics:** How does a VM or cluster get its token without a password? It queries an internal non-routable link local metadata endpoint: <http://169.254.169.254/metadata/identity/oauth2/token>.
3. **Data Plane vs. Control Plane Authorization:** Having a Managed Identity is only half the battle. You must explicitly grant that identity an RBAC role on the target resource (e.g., giving Databricks' Managed Identity Storage Blob Data Reader on ADLS).

7. IAM & Role-Based Access Control (RBAC)

IAM (Identity & Access Management) is the overall security desk, and **RBAC** is the rulebook. Instead of setting custom permissions for individual employees, you assign them standard job badges (roles) like Reader, Contributor, or Owner.

1. **The Owner / Control Plane Trap:** Being an Owner or Contributor at the Resource Group level gives you control over the *infrastructure* (Control Plane), but **does not automatically allow you to view data inside the storage account** (Data Plane). To read files in ADLS, you specifically need a Data Plane role like Storage Blob Data Reader.
2. **Inheritance & Deny Assignments:** Permissions inherit downward through the scope chain. Standard Azure RBAC is **additive** (allows accumulate). Explicit Deny Assignments exist, but they are rare and applied automatically by Azure Blueprints or Managed Applications, taking precedence over Allow.
3. **Least Privilege Principle:** Assigning permissions at the Subscription scope is a common security failure. Roles should be assigned at the lowest necessary level (Resource or Resource Group).

8. Azure Key Vault (AKV)

Azure Key Vault is a ultra-secure cloud digital safe. It stores three main asset types:

- **Secrets:** API keys, database connection strings, passwords.
 - **Keys:** Cryptographic encryption keys (RSA/HSM) for data encryption.
 - **Certificates:** SSL/TLS public key certificates for domain identity.
1. **Access Permission Models (Legacy vs. Modern):** Key Vault supports two conflicting permission models:
 - **Vault Access Policies (Legacy):** Simple access list defined directly on the vault.
 - **Azure RBAC (Modern & Recommended):** Granular roles like Key Vault Secrets User. Mixing these causes permission confusion.
 2. **Soft Delete & Purge Protection:** When a Key Vault or secret is deleted, Azure keeps it in a "soft-deleted" recoverable state by default for up to 90 days. Enabling **Purge Protection** prevents anyone (even subscription Owners!) from permanently deleting a secret before its retention period ends.
 3. **Key Vault Secret Versioning:** When you update a secret's value, Key Vault creates a new version while preserving older versions. Applications referencing secrets should fetch without specifying a version ID unless a static version lock is required.

9. Azure Data Factory (ADF)

Azure Data Factory is a cloud-based data integration and orchestration engine (ETL/ELT). Imagine ADF as a **Traffic Controller** or **Factory Floor Manager**. It doesn't store data itself; instead, it coordinates pulling data from Source A, transforming it using external compute engines (like Databricks or SQL), and loading it into Destination B.

1. **Integration Runtimes (IR) Types:** The compute engine ADF uses to execute activities come in 3 flavors:
 - **Azure IR:** Fully managed cloud compute for cloud-to-cloud transfers.
 - **Self-Hosted IR (SHIR):** Required to connect to **on-premises corporate databases** behind internal firewalls.
 - **Azure-SSIS IR:** Used to run legacy SQL Server Integration Services packages natively in Azure.
2. **ADF Pipeline Triggers:**
 - **Schedule Trigger:** Executes at fixed wall-clock times (e.g., every day at 8 AM).
 - **Tumbling Window Trigger:** Evaluates historic contiguous time intervals (ideal for processing backlogged historical partitions).
 - **Storage Event Trigger:** Executes dynamically when a new file lands in ADLS/Blob storage.

3. **Linked Service vs. Dataset:** A *Linked Service* is purely the connection string/authentication method to a service. A *Dataset* is a specific structure or file path pointing to data using that Linked Service.

10. Azure Networking Essentials (VNet, Private Endpoints & Service Endpoints)

By default, standard Azure PaaS services (Storage, SQL, Synapse) have endpoints exposed to the public internet (secured by firewalls). **Azure Networking** allows you to put resources inside a private **Virtual Network (VNet)**—like building a high-security wall and private highway system around your infrastructure.

1. **Private Endpoints vs. Service Endpoints:**
 - **Service Endpoint:** Optimizes traffic routing over the Microsoft Azure backbone, but the target resource **still retains a public IP address**.
 - **Private Endpoint:** Creates a dedicated private Network Interface (NIC) inside your VNet with a **private IP address** (e.g., 10.0.0.4). Public internet access to the resource can be completely disabled.
2. **Private DNS Zone Resolution Trap:** When using Private Endpoints, the public FQDN (e.g., mystorage.blob.core.windows.net) must resolve to the *private IP address* via an **Azure Private DNS Zone** (privatelink.blob.core.windows.net). Misconfigured DNS zones are the #1 cause of deployment failure in private networks.

11. Azure Databricks

Azure Databricks is an enterprise-grade Apache Spark processing engine optimized for cloud analytics, machine learning, and data engineering. It allows teams to process massive datasets collaboratively using interactive notebooks.

1. **Double Billing Structure:** Running a Databricks cluster charges you twice:
 - Azure Infrastructure Cost (for underlying Virtual Machines running Spark nodes).
 - Databricks DBU Cost (Databricks Units charged for software usage).
2. **Control Plane vs. Data Plane Architecture:** Databricks UI and web application sit in a Microsoft-managed **Control Plane**. The actual worker nodes (VMs) and data processing run inside *your* subscription's Virtual Network (**Data Plane / Customer VNet**).
3. **Auto-Termination Safety:** Forgotten running clusters can drain thousands of dollars overnight. Always configure **Auto-Termination** (e.g., shut down after 20 minutes of inactivity).
4. **Delta Lake Time Travel:** Databricks uses Delta Lake by default, which keeps transaction

logs (_delta_log). You can query historical versions of a dataset (SELECT * FROM table TIMESTAMP AS OF ...), but raw files aren't deleted unless you execute a VACUUM command.

dsiu/';p.

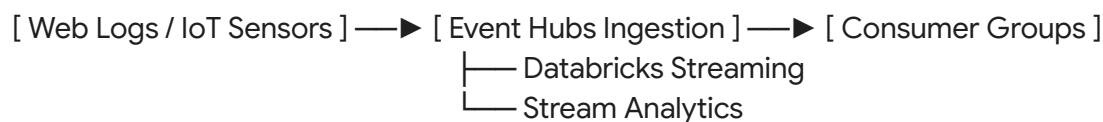
12. Azure Synapse Analytics

Azure Synapse Analytics is a unified analytics ecosystem. Rather than buying separate tools for ETL pipelines, data warehousing, and file querying, Synapse combines all of them inside a single interface called **Synapse Studio**

1. **Serverless SQL Cost Trap:** Serverless SQL charges based on **data scanned per query** (approx. \$5 per TB scanned). Running SELECT * on uncompressed Parquet/CSV files across terabytes of data can trigger unexpected billing spikes. *Best practice:* Use partition pruning and file formats like Parquet.
2. **Dedicated SQL Distribution Strategies:** Dedicated SQL Pools (Data Warehouses) store data across 60 internal databases using 3 distribution patterns:
 - **Hash Distribution:** Optimizes large fact tables by hashing a specific column.
 - **Round Robin:** Default distribution for staging tables; splits data evenly.
 - **Replicated:** Copies small dimension tables across all compute nodes for fast joins.
3. **Synapse Pipelines vs. Azure Data Factory (ADF):** Synapse Pipelines are built on the exact same engine as ADF, but ADF remains a standalone service with earlier feature releases and independent lifecycle management.

13. Azure Event Hubs & Real-Time Analytics

Azure Event Hubs is a real-time data ingestion service capable of processing millions of events per second. Imagine Event Hubs as an **Airport Luggage Carousel** or a **Firehose Buffer**. When thousands of IoT devices, web apps, or mobile devices emit logs simultaneously, Event Hubs buffers the incoming stream so consumer engines (like Databricks, Stream Analytics, or Functions) can process it without crashing.



1. **Partitions & Ordering Guarantee:** Event Hubs splits data into multiple parallel streams called **Partitions**. Strict message ordering is guaranteed *only within a single partition*. If event sequence matters, events must share the same partition key.
2. **Consumer Groups & Reader Decoupling:** Consumer Groups allow multiple independent applications (e.g., real-time fraud monitoring AND historical data storage) to read the

same event stream at their own pace without interfering with each other.

3. **Event Hubs Capture:** A zero-code built-in feature that automatically dumps real-time streaming data directly into ADLS Gen2/Blob storage in Parquet or Avro format for long-term storage.

14. End-to-End Enterprise Architecture

The diagram below demonstrates how these components interact securely in a real-world enterprise pipeline:

